

Multiple Choice (10 marks)

1. In adolescence:
 - (a) Brain growth is faster than at any other stage in life
 - (b) Peak bone mass occurs before the growth spurt ends
 - (c) Girls physiologically increase their percentage body fat
 - (d) Type 2 obesity related diabetes remains a very rare problem
2. Supplementation with long-chain fatty acids EPA and DHA during pregnancy and infancy will likely benefit the following:
 - (a) Visual performance
 - (b) Motor performance
 - (c) Behaviour disorders
 - (d) Memory
3. Which one of these organs has the lowest energy expenditure per unit of body mass (kg)?
 - (a) Brain
 - (b) Kidney
 - (c) Liver
 - (d) Skeletal Muscle
4. It has been observed that a reduction in family size and improved sanitary condition have led to an increase in allergic conditions. The 'hygiene hypothesis' suggests that
 - (a) The frequent use of antiseptic swipes will reduce allergies
 - (b) Living on a working farm will help reducing the development of atopic diseases
 - (c) Particular care must be taken during breast feeding in high risk infants
 - (d) Living on a working farm will increase the development of atopic diseases
5. Bone mineral accretion
 - (a) Is rapid in the first 10 weeks of foetal life
 - (b) Is slow in infancy
 - (c) In childhood, adolescence and young adulthood predicts risk of fracture in old age
 - (d) Is particularly high after the menopause in women

True / False (10 marks)

Answer True or False

6. True or False: Infants born to mothers who are vegan are at increased risk of deficiency in Vitamin C.
7. True or False: The concentration of a substance in food is the only factor needed to assess its safety for consumers.
8. True or False: The safety of pesticide chemicals, substances from food packaging, and color additives must be shown before they can be used in food.
9. True or False: Muscle glycogen is the primary fuel source for muscle contraction during short-term intense activities like sprinting.
10. True or False: In type 2 diabetes, impaired insulin secretion is not a factor in causing hyperglycemia.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the Mediterranean diet and analyze how its components may contribute to preventing age-related cognitive decline.
12. Discuss the role of plant sources of amino acids in human nutrition, focusing on how they provide essential amino acids and the implications of varying amino acid profiles.

End of Paper